

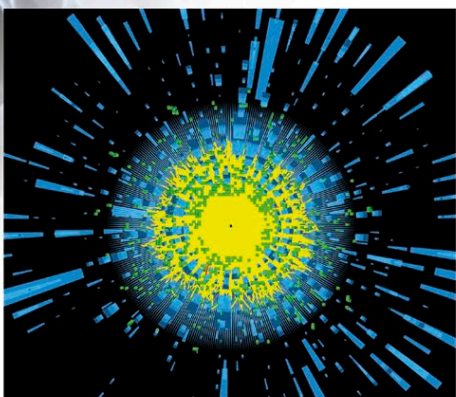
PARTICLE PHYSICS

For decades, physicists have sought a better understanding of matter and the fundamental forces. Research has centered on smashing particles together in devices called particle accelerators. These experiments have identified hundreds of different particles (most of them highly unstable), which vary in properties such as mass, electric charge, and “spin.” Known particles and their interactions are currently explained by a theory called the standard model of particle physics, which includes a particle classification (see table, right). One recently discovered addition to the model is the Higgs boson. This is the force carrier for an energy field that permeates space and endows other particles with mass through its interactions with them. However, the hypothetical particle called the graviton—proposed to be a force carrier for gravity—does not fit comfortably into the model. This is because the best theory of gravity

(general relativity, see pp.42–43) is incompatible with aspects of the standard model. New theories such as string theory (see panel, below) attempt to unite gravity with particle physics.

COLLISION EVENT

The results of a collision between xenon ions in the Large Hadron Collider are shown here as a computer graphic. The blue tracks represent baryon interactions.



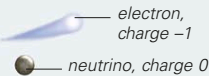
CLASSIFICATION OF PARTICLES

Physicists distinguish composite particles, which have an internal structure, from elementary particles, which do not. They also divide particles into fermions and bosons. Fermions (leptons, quarks, and baryons) are the building blocks of matter. Bosons (gauge bosons and mesons) are primarily force-carrier particles.

ELEMENTARY PARTICLES

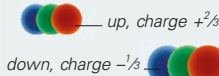
Leptons and quarks form matter, while gauge bosons carry forces. Quarks feel the strong nuclear force, but leptons do not.

LEPTONS



Six different leptons exist, but the 2 above are the only stable ones and are those that occur in ordinary matter.

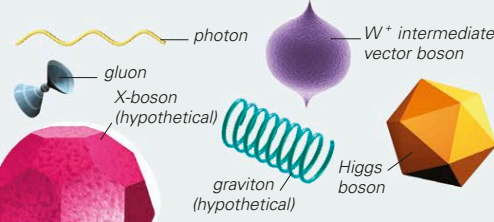
QUARKS



There are 6 “flavors” of quark, but only 2 occur in ordinary matter: “up” and “down.” Each can exist in any of 3 “colors.”

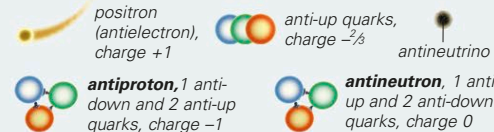
GAUGE BOSONS

These are force-carrier particles. Some shown are hypothetical.



ANTIPARTICLES

Most particles have an antimatter equivalent that has the same mass but whose charge and other properties are opposite.

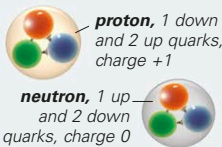


COMPOSITE PARTICLES

Also known as hadrons, these are composed of quarks, antiquarks, or both, bound by gluons.

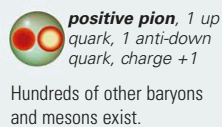
BARYONS

Relatively large-mass particles containing 3 quarks.



MESONS

Particles containing a quark and an antiquark.

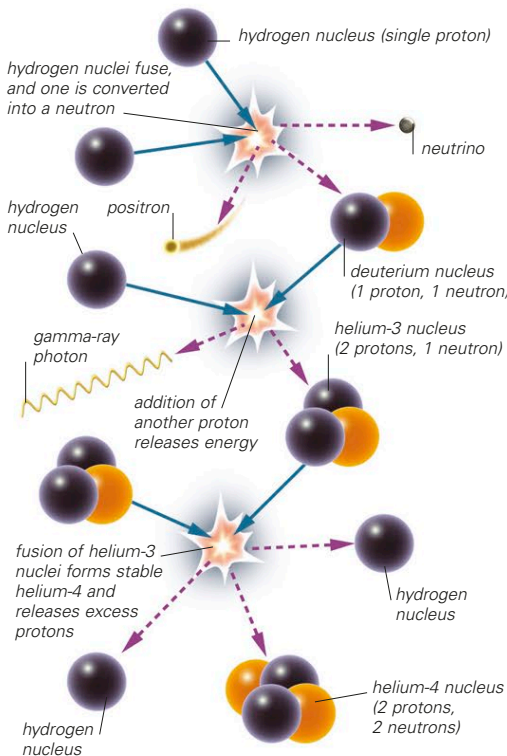
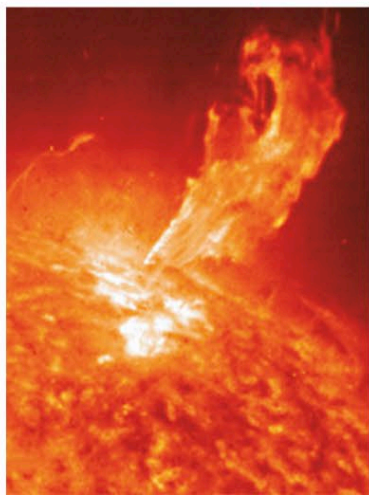


EXOTIC PARTICLES

Further particles have been hypothesized that do not have a place in this particle classification. They include magnetic monopoles and WIMPs (weakly interacting massive particles).

NUCLEAR FISSION AND FUSION

Twentieth-century physicists learned that atomic nuclei are not immutable but can break up or join together. In nature, unstable atomic nuclei can spontaneously disassemble, giving off particles and energy, measured as radioactivity. Similarly, in the artificial process of nuclear fission, large nuclei are intentionally split into smaller parts, with huge energy release. On a cosmic scale, a more important phenomenon is nuclear fusion. In this process, atomic nuclei join, forming a larger nucleus and releasing energy. Fusion powers stars and has created most or all atoms of the majority of naturally occurring chemical elements. The most common fusion reaction in stars joins hydrogen nuclei (protons) into helium nuclei. In this and other fusion reactions, the products of the reaction have a slightly lower mass than the combined mass of the reactants. The lost mass converts into huge amounts of energy, in accordance with Einstein’s famous equation $E = mc^2$ that links energy (E), mass (m), and the speed of light (c) (see p.41).



FUSION REACTION IN THE SUN

In stars the size of the Sun or smaller, the dominant energy-producing fusion process is called the proton-proton chain. This chain of high-energy collisions fuses hydrogen nuclei (free protons), via several intermediate stages, into helium-4 nuclei. Energy is released in the form of gamma-ray photons and in the movement energy of the helium nuclei. Positrons and neutrinos are also produced.

EXPLORING SPACE

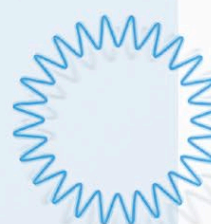
STRING THEORY

For decades, physicists have sought a “theory of everything” that will unify the four fundamental forces of nature and provide an underlying scheme for how particles are constructed. A leading contender is string theory, which proposes that each elementary particle consists of a tiny vibrating filament called a string. The vibrational modes, or frequencies, of these strings lend particles their varied properties. Although it sounds bizarre, many leading physicists are enthusiastic adherents of string theory.

LOW-FREQUENCY STRING

VIBRATING STRINGS

A string is closed, like a loop, or open, like a hair. The two closed strings shown here are vibrating at different resonant frequencies, just as the strings on a guitar have rates at which they prefer to vibrate.



HIGH-FREQUENCY STRING